

# Skills Summary

## Sorting Shapes by What Makes Them That Shape

*Use this reference while completing your worksheet or when studying.*

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### Skill 1 — Counting Sides and Corners

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A **side** is a straight edge. A **corner** is the point where two sides meet.

**In every flat shape with straight sides, the number of sides equals the number of corners.** Triangle: three sides, three corners. Square and rectangle: four and four. Hexagon: six and six. Circle: no straight sides and no corners — one curved edge all the way round.

**How to count without losing your place:** start at one corner, touch each side once, and stop when you get back to where you started.

**Example:** Which has more straight sides, a rectangle or a triangle?

**Step 1.** The question asks about sides, so count the straight edges of each shape and then compare the two numbers.

**Step 2.** Rectangle: top, bottom, left, right — that is four. Triangle: three.

⇒  $4 > 3$ , so the rectangle has more.

**Try it:** Which has more straight sides, a hexagon or a triangle?

check: 6, the hexagon,  $6 > 3$

### Skill 2 — Colour and Size Do Not Matter

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**Defining attributes** are the things you can count or check: number of straight sides, number of corners, whether the shape is closed, whether the edges are straight or curved. These decide what the shape IS.

**Non-defining attributes** are colour, size, and which way the shape is turned. A shape keeps its name when you make it bigger, paint it another colour, or stand it on a point.

**Test to use:** if you are not sure, count the sides. If the counts match, the shapes have the same name.

**Example:** A tiny triangle and a huge triangle — are they both triangles?

**Step 1.** Size is not something you count on the shape itself, so ignore it and count the sides on each shape instead.

**Step 2.** Tiny one: three sides. Huge one: three sides.

⇒  $3 = 3$ , so yes — both are triangles.

**Try it:** A red square and a blue square — count the corners of each. Are they both squares?

check: 4 = 4, yes

### Skill 3 — Sorting Shapes into Groups

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**Sorting means picking one rule and using only that rule.** Good rules to sort by: “has four straight sides”, “has three corners”, “has no straight sides”.

**Bad rules:** “is red”, “is big”, “looks like a box”. Those change from shape to shape without changing what the shape is.

**After sorting, count each group** — then you can say which group has more.

**Example:** Sort a square, a rectangle, a trapezoid, a triangle and a circle by “has four straight sides”. Which group is bigger?

**Step 1.** Pick the one rule and test each shape against it, then count how many landed in each group.

**Step 2.** Four straight sides: square, rectangle, trapezoid — that is three. Not four sides: triangle, circle — that is two.

$\Rightarrow 3 > 2$ , so the four-sided group is bigger.

**Try it:** Out of six shapes, two have three corners and the rest do not. Which group is bigger?

check: 2 < 4, the “not three corners” group

### Skill 4 — Putting Shapes in Order

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**To put shapes in order, turn each shape into a number first** — how many sides, or how many corners — and then order the numbers.

**Fewest to most** means smallest number first. **Most to fewest** means largest number first.

**Remember the circle:** it has zero straight sides, so it comes first in a fewest-to-most list and last in a most-to-fewest list.

**Example:** Put a square, a triangle and a hexagon in order from the fewest corners to the most.

**Step 1.** Order words like “fewest” are about numbers, so count each shape first and then sort the numbers, not the pictures.

**Step 2.** Square: four. Triangle: three. Hexagon: six. Smallest first: three, then four, then six.

$\Rightarrow 3, 4, 6$

**Try it:** Put a pentagon, a triangle and a square in order from the fewest corners to the most.

check: 3, 4, 5